Assignment # 1

Team Assignment

Introduction to Excel Modeling

Expectations:

1. Complete reading the Excel handout material (Introduction to Excel Part-I), then **make a list of the tools in Excel you found to be useful** (at least **ten tools**).
2. Submit a technical report to describe the overall modeling solution process to solve the attached External Problem.
3. Remember to use the sandwich structure (Introduction, main body, discussion, ...) and follow the necessary format for all modeling assignments (see the First day materials and the Presentation of Technical Report Expectations). The main body ***"model"*** should **at least** include:
   - Problem definition (this includes the given information and data as well as the requirements and objectives of the problem),
   - Assumptions,
   - Heuristics (***NA***),
   - Modeling steps and description (equations, algorithms, sketches, etc.),
   - Model implementation and results (calculations, excel sheet, numbers, graphs, resulting relations, etc.),
   - Result discussion and evaluation (interpretations, justifications, comparisons, validation, calibration, etc.).
4. Attach the assignment checklist on the top of the report cover page. Go through **all requirements** mentioned in the checklist to ensure that you have completed the assignment properly. Write down any exciting features you may have completed on your own in the relevant item of the checklist.

External Problem

An industrial equipment supplier manufactures a custom composite unit consisting of a solid cylindrical body capped by a cone end, as shown in Figure 1.

The diameter D_o of the cylinder varies from 80 mm to 200 mm in uniform increments of 10 mm. For a given diameter D_o, the remaining geometric parameters are defined as

$L_{Cone} = 1.25D_o$

$L_{Cylinder} = 3D_o$

The batch size is 150 units for each distinct value of D_o. The components are fabricated from an aluminum alloy with density $\rho = 2.70$ g/cm^3.

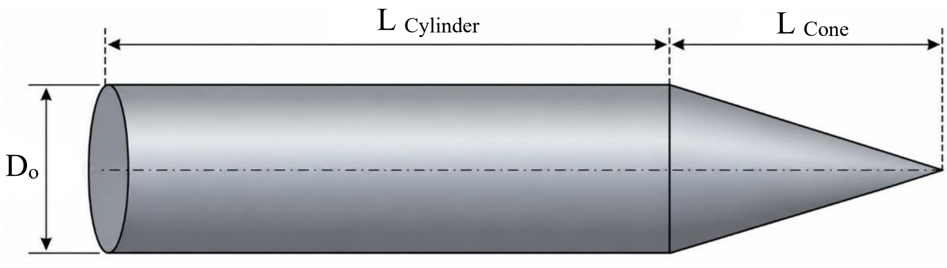

Figure 1: Unit dimensions.

Requirements:

I- Finished Component Mass: Calculate the total mass (in kg) of aluminum required to produce the batch of 150 finished units for each corresponding D_o.

II- Machining Scrap Percentage: Prior to final machining, raw castings are produced with dimensions oversized by 4 mm on all diameters and lengths (D_{raw}) = D_o +4 mm, $L_{raw\ Cylinder} = L_{Cylinder}$ +4 mm, $L_{raw\ Cone} = L_{Cone}$ +4 mm. Determine the mass percentage of material lost to scrap during machining for each R_o:

$$Scrap\ \% = \left(\frac{M_{raw} - M_{finished}}{M_{raw}}\right) \times 100$$

III- Packaging Fill Mass: Each finished unit is packaged individually inside a rectangular cuboid container with dimensions 4.4 D_o × 1.2 D_o × 1.2 D_o. The remaining void space inside the box is filled with expanding foam sealant (ρ_{foam} = 0.03 g/cm³). Calculate the total mass of foam (in g) required per unit for each D_o.

IV- Surface Coating and Finishing Cost: The full external surface area of each unit is sprayed with a protective polymer coating of thickness 0.2 mm. Coating material cost: 6 SAR/mL. Processing labor cost: 15 SAR/kg of coated items. Determine the total coating and labor cost (in SAR) per unit for each D_o.

V- Graphical Representation: Plot your results as a function of outer radius D_o (in cm) on two separate graphs:

- **Graph 1**: One unit mass in kg (Task I) and unit foam mass in g (Task III) vs. D_o.
- **Graph 2**: Machining scrap percentage (Task II) and total finishing cost per unit (Task IV) vs. D_o.

Submission

- Each team should submit: (**1**- The report as a Word file, **2**- The report with the checklist and team process check as a pdf file, **3**- An Excel file for the model and its solution).
- Each file should be named like (A? - G? - T?).